AI-DRIVEN LEARNING WORKSPACE

StudyMapper

AI-Driven Personalized Learning & Academic Management System

PROJECT OVERVIEW

A local-first study workspace that turns a syllabus into an adaptive plan, generates notes and practice, and scores weakness from activity.

React 19 · Dexie / IndexedDB · OpenRouter

Plan → Learn → Practise → Improve

the same records show what to revisit

Live demo · study-mapper.vercel.app

TEAM — 5 MEMBERS

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Batch 4 · Department of CSE

GUIDE

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DEPARTMENT & INSTITUTION

Computer Science & Engineering

Anantha Lakshmi Institute of Technology & Sciences (Autonomous)

Thirteen sections, four moves

Frame the problem, design the system, prove it, then close.

SET THE STAGE 01–04

01 Title
project, team, guide

02 Agenda
the route through 13

03 Abstract & Motivation
why it exists

04 Problem Statement
what is broken

Why does it exist?

AIM & DESIGN 05–07

05 Project Objectives
what it must do

06 Literature Survey
what others miss

07 System Architecture
how the parts connect

What should it do?

BUILD & PROVE 08–10

08 Methodology & Algorithm
how it decides

09 Implementation Details
what it runs on

10 Results & Analysis
what is verified

How is it built and checked?

CLOSE 11–13

11 Conclusion
what it contributes

12 Future Scope
what comes next

13 References
what it rests on

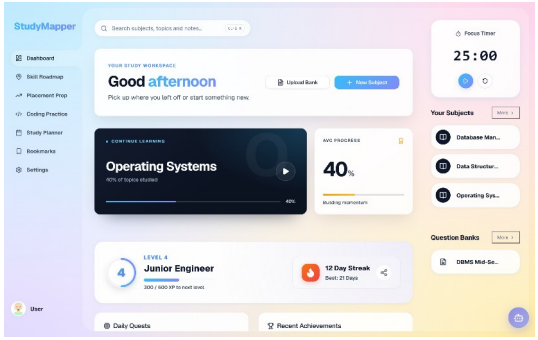
What does it mean?

About twelve minutes, then questions. Every slide has one job.

From syllabus to next action

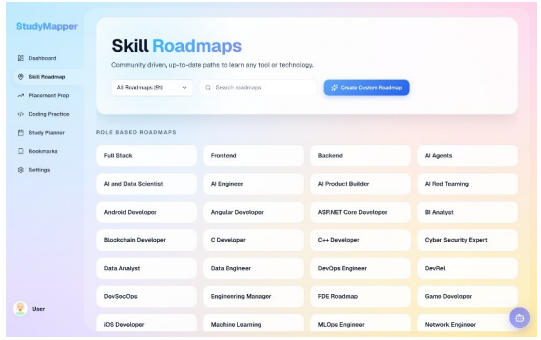
Four screens from the running build — the same records feed every one of them.

RUNNING BUILD · 1440 × 900



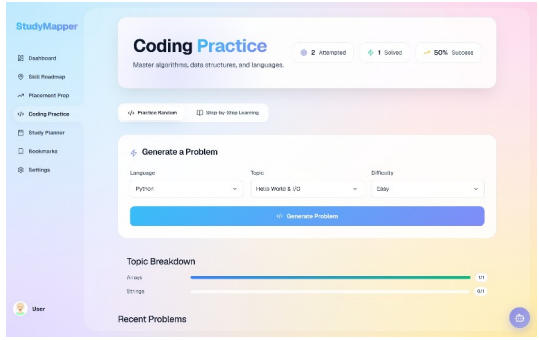
Dashboard

subjects + question banks



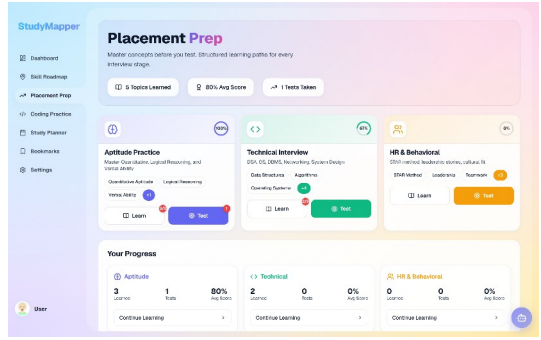
Skill roadmap

91 roadmaps, progress saved



Coding practice

Python runs in the browser



Placement prep

aptitude, technical, HR

THE QUESTION

Can a student's own activity produce the next study plan?

- **One workspace**

  Notes, deadlines, tests and progress stop living in different apps.

- **Evidence, not guesswork**

  Study activity updates a weakness score that shows what to revisit next.

- **Works offline**

  The core loop runs in the browser; AI and sign-in are optional.

The gap is response, not content

Students do not lack material. They lack a system that reacts to their week.

01 **Fragmented tools** — 4 apps

Notes, deadlines, tests and progress live in different apps, so context is rebuilt every day.

02 **Static plans** — 1 plan

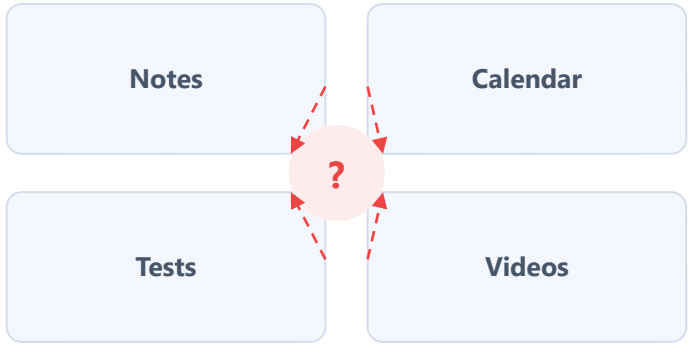
One plan is made once and survives every missed week, whatever the topic's difficulty.

03 **Late feedback** — exam day

Mistakes surface at the exam, not during practice, so the next study session cannot respond.

TODAY'S STUDY WEEK

four tools, no shared record



No shared record — context is lost at every switch.

4 tools · 0 shared records

the week cannot compound

THE CONSEQUENCE

The same hours are spent on every topic, and the plan never learns from the week.

Make the whole week one system

Four goals, one test: a student can run a full study week inside one app.

01

A plan that recomputes

Build the schedule from the exam date, topic difficulty and available days — then compress it in panic mode.

exam date + difficulty + panic mode

02

Material on demand

Generate notes, diagrams, question banks and model answers that the student can edit, not opaque output.

editable, source-linked

03

Practice that measures

Turn stored study events into a 0–100 weakness score rather than a single mark at the end.

events → weakness

04

One local record

Keep the study loop in the browser: no server required, AI optional, identity optional.

Dexie · offline

IN SCOPE

what this build actually does

- Adaptive planner
- Generated notes & questions
- Mock exams
- Coding practice
- Weakness analytics
- Offline study

One test: a full study week without leaving the app.

OUT OF SCOPE

scope decisions, not missing pieces

- Institutional timetabling
- Proctored exams
- Teacher dashboard
- Native mobile app
- Cross-device sync

Five approaches, one missing loop

Each category solves part of the problem. None closes the loop for one student.

01 Adaptive platforms

DOES WELL

Sequence content by mastery and difficulty.

MISSES

Rarely react to one student's week.

02 AI tutors

DOES WELL

Generate explanations and practice on demand.

MISSES

Cloud-only, disconnected from stored work.

03 LMS analytics

DOES WELL

Track cohort performance and completion.

MISSES

Serve the institution, not the individual plan.

04 Spaced repetition

DOES WELL

Schedule review at the edge of recall.

MISSES

Isolated from syllabus, notes and practice.

05 Gamified apps

DOES WELL

Sustain daily habit with streaks and rewards.

MISSES

Reward activity, not understanding.

THE GAP

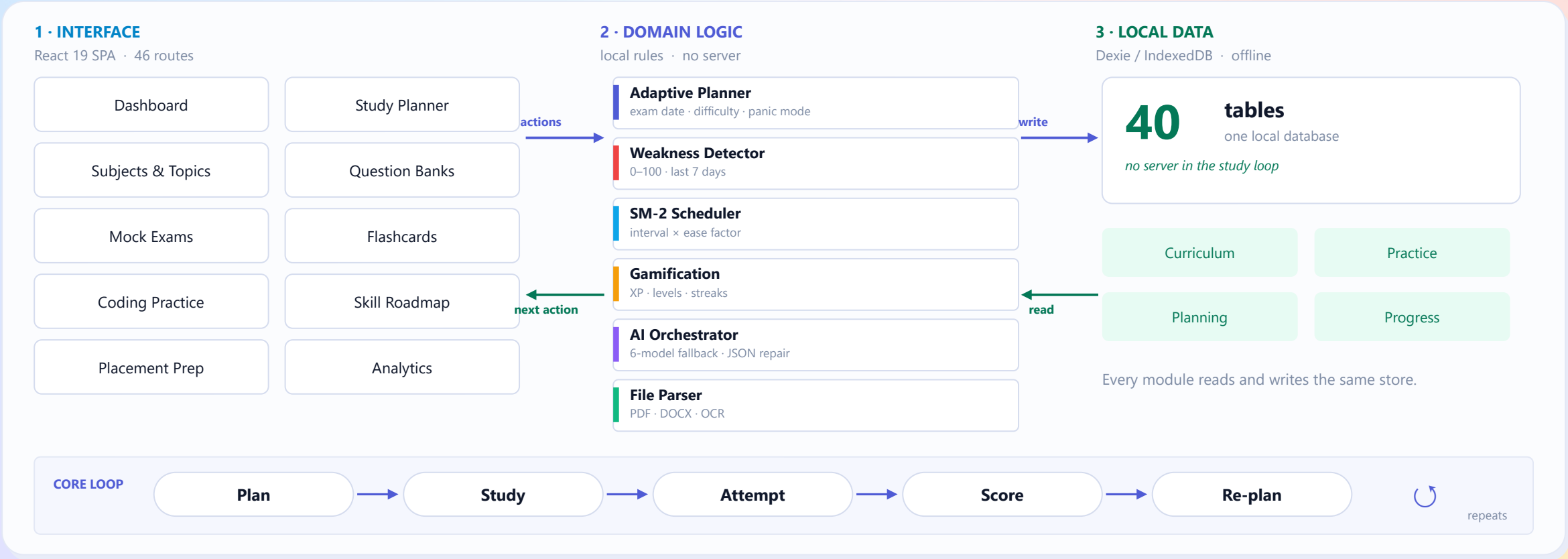
No category combines planning, generation, measurement and offline study for one learner.

that is the space StudyMapper fills

Design comparison, not a systematic review. No effectiveness claim is attributed to any category.

Every screen, one local core

The interface, the rules and the record — with AI and identity as optional edges.



OPTIONAL / EXTERNAL

- OpenRouter · 6-model fallback
- Firebase Auth · optional
- Pyodide · Python in browser
- Mermaid + KaTeX

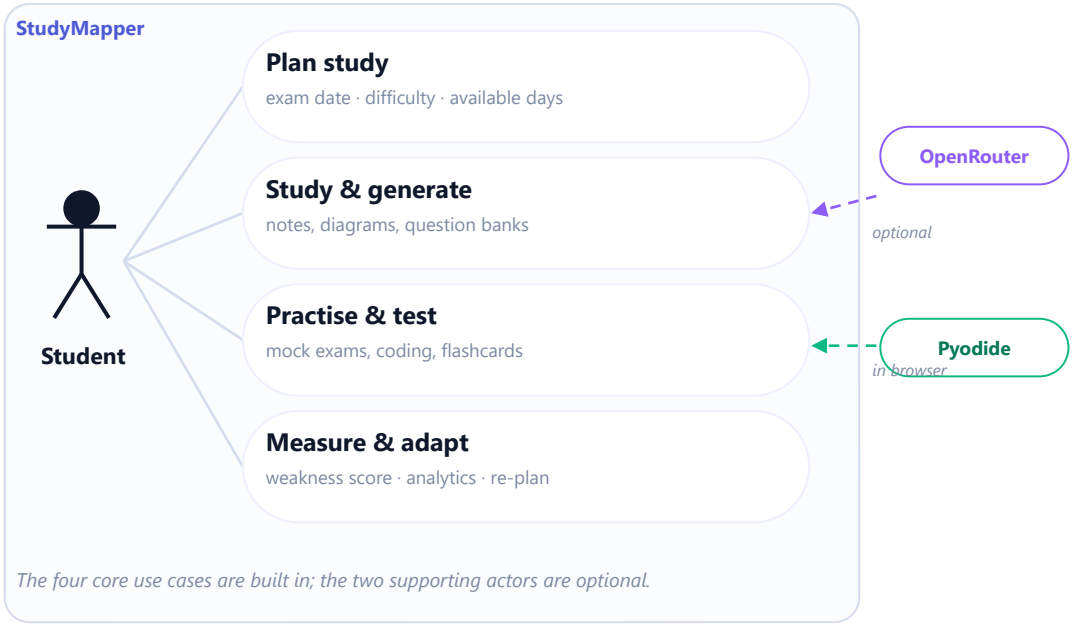
dashed = network

How a study action becomes the next plan

A use-case view of what the student can do, and an activity view of what the system does with it.

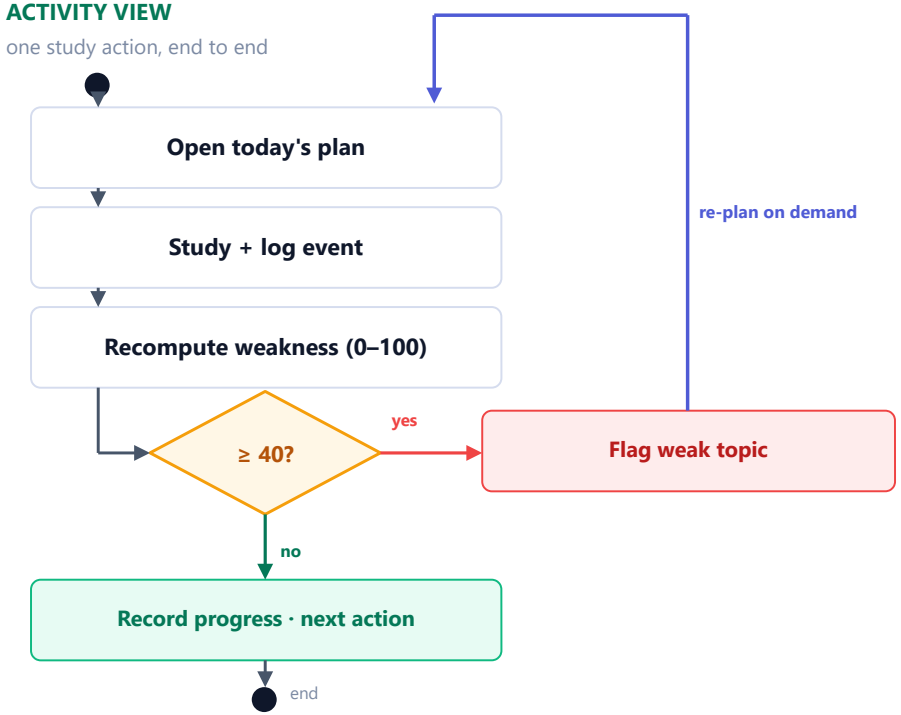
USE-CASE VIEW

actor · system boundary · optional actors



ACTIVITY VIEW

one study action, end to end



WEAKNESS SCORE · 0–100 · LAST 7 DAYS

+30 failed code · +0.5 × (100 − accuracy) · +10 pomodoro · −20 code pass · −10 notes · flag at ≥ 40

SM-2 SCHEDULER

interval ← interval × ease factor

the queue is built from topic and history

A browser app, not a server stack

Every choice removes something to install, host or pay for.

UI

Interface

What the student sees and touches.

- **React 19**
  46 routed views
- **Vite**
  6.21 s production build
- **React Router**
  route-level structure
- **CSS tokens**
  one palette, light + dark

AI

Intelligence

Generation and in-browser computation.

- **OpenRouter**
  six-model fallback
- **Pyodide**
  Python in the browser
- **Mermaid + KaTeX**
  diagrams and maths
- **Tesseract · pdf.js · Mammoth**
  PDF, DOCX, OCR

DB

Data

Where the study record actually lives.

- **Dexie.js**
  IndexedDB wrapper
- **40 tables**
  curriculum to progress
- **Firebase Auth**
  optional identity
- **Offline**
  the core loop works

QA

Quality

How the build is checked and shown.

- **Vitest**
  47 tests in 14 files
- **Vite build**
  production bundle
- **Puppeteer**
  19 interface captures
- **Vercel**
  live demo

RUNS ON

Any modern browser.

No GPU · no server · no model hosting · no paid tier

VERIFIED BY

47/47 tests · 6.21 s build · 19 captures

the same build on a laptop and the live demo

What is verified — and what is not

Functional completeness and build health, measured. Learning gains are the next study.

47/47
tests passing

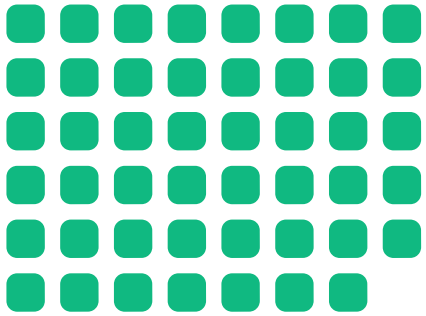
6.21 s
production build

40
database tables

46
routed views

TEST WALL

47 tests · 14 files · 0 failures

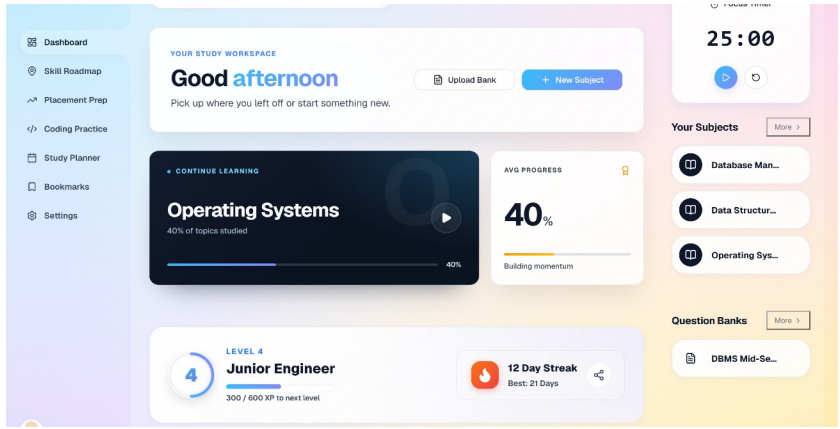


47 / 47 passing, no skips

BEFORE → WITH STUDYMAPPER

workflow comparison, not an experiment

Four apps	→ **One workspace**
Plan made once	→ **Plan recomputed**
Exam feedback	→ **Practice feedback**



Dashboard
progress, level and streak

NOT CLAIMED Learning gains · institutional deployment · teacher dashboard · native app · cross-device sync

A complete, working learning loop

What the project contributes — stated as mechanisms, not adjectives.

ADAPTIVE

The plan is an output, not a document.

Exam date, topic difficulty and panic mode recompute the schedule on demand.

EXPLAINABLE

Every flag traces to a stored record.

A 0–100 score over seven days, a threshold of 40 and fixed, documented weights.

LOCAL-FIRST

Study survives a bad connection.

Notes, plans and attempts live in Dexie; AI and sign-in are optional.

WHAT THE PROJECT CONTRIBUTES

A closed loop, not a bundle
Planning, content, practice and measurement share one data model, so evidence shows what to revisit.

AI with a safety net
Six-model fallback, streaming and JSON repair; generated material stays editable.

Local-first by design
The core loop runs offline; identity is never required for study.

EVIDENCE AT A GLANCE

47/47 tests	**6.21 s** build
40 tables	**46** views

Live demo · study-mapper.vercel.app

The loop is built and verified. Whether it improves learning is the experiment ahead.

Evidence first, then scale

The next step is a classroom study; the roadmap follows from what that study reveals.

NOW → 3 MONTHS

Classroom pilot
Pre/post quiz with a real cohort — the first test of learning impact.

Optional sync
Account-based, conflict-safe sync without losing local speed.

Accessibility pass
Keyboard paths, contrast and screen-reader labels.

HOW WE WILL KNOW

pre/post gain in one cohort

NEXT YEAR

Native mobile shell
Capture small study sessions between classes.

Mentor view
A cohort view for teachers — aggregate, not surveillance.

Content review queue
Check AI-generated material before sharing.

HOW WE WILL KNOW

weekly active students · review turnaround

BEYOND

Multi-language content
Localise notes and questions, not just the interface.

LMS integration
LTI-style hand-off to institutional course spaces.

Longitudinal study
A multi-semester evaluation with an education department.

HOW WE WILL KNOW

multi-semester retention and outcomes

Prove the loop helps a student learn before adding surface area.

Sources used to verify this work

Primary sources first; technical documentation only where it was needed to confirm a claim.

PRIMARY & PLATFORM

1 P. Venubhaskar et al., "StudyMapper: AI-Driven Personalized Learning and Academic Management System", project abstract, ALITS, 2026–27.

2 StudyMapper source repository, GitHub — github.com/palturuvenubhaskar/StudyMapper. Accessed 28 October 2026.

3 StudyMapper live application — study-mapper.vercel.app. Accessed 28 October 2026.

4 Meta Open Source, React 19 documentation — react.dev. Accessed 28 October 2026.

5 Vite documentation — vite.dev. Build tooling and the production bundle.

6 Dexie.js documentation — dexie.org. IndexedDB schema, transactions and live queries.

7 Google, Firebase Authentication documentation — firebase.google.com/docs/auth.

LIBRARIES & METHODS

8 OpenRouter API documentation — openrouter.ai/docs. Model routing, streaming and fallback.

9 Pyodide documentation — pyodide.org. Python runtime in the browser.

10 Tesseract.js, pdf.js and Mammoth.js documentation. PDF, DOCX and OCR parsing.

11 Mermaid — mermaid.js.org; KaTeX — katex.org. Diagrams and mathematical notation.

12 P. A. Woźniak, "SuperMemo 2: Algorithm and implementation", 1987. Basis of the SM-2 scheduler.

13 Anki manual — docs.ankiweb.net. Reference implementation of spaced repetition.

14 Vitest documentation — vitest.dev. Unit testing for Vite.

All web sources accessed October 2026 · Repository: github.com/palturuvenubhaskar/StudyMapper · Live: study-mapper.vercel.app